Chapter 1.5 Properties of the Integral

Rick Durrett Probability: Theory and Examples

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Built from the local Chapter 1 LLMwiki flash cards.

Roadmap for 1.5

Learning goal

Use Jensen, Holder, Minkowski, Fatou, MCT, DCT, and bounded convergence with exact hypotheses.

1. Inequalities: Jensen, Holder, Minkowski.
2. Limit theorems: bounded convergence, Fatou, MCT, DCT.
3. Exercise patterns: endpoint Holder, $L^p \to L^\infty$, absolute continuity, dense simple functions, series exchange.

Jensen's Inequality

Theorem

If μ is a probability measure, φ is convex, and the integrals are defined, then

$$\varphi\left(\int f \, d\mu\right) \leq \int \varphi(f) \, d\mu.$$

LLMwiki: `Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral`

Holder's Inequality

Theorem

If $1 < p, q < \infty$ and $p^{-1} + q^{-1} = 1$, then

$$\int |fg| \, d\mu \le \|f\|_p \, \|g\|_q \, .$$

LLMwiki: `Holder's inequality; ID: durrett_ch1_holder_integral`

Minkowski's Inequality

Theorem

For $1 \leq p \leq \infty$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p .$$

LLMwiki: `Minkowski's inequality; ID: durrett_ch1_minkowski`

Bounded Convergence

Theorem

On a finite-measure space, if $f_n \to f$ pointwise and $|f_n| \leq M$, then

$$\int f_n \, d\mu \to \int f \, d\mu.$$

LLMwiki: `Bounded convergence theorem; ID: durrett_ch1_bounded_convergence`

Fatou's Lemma

Theorem

If $f_n \geq 0$, then
$$\int \liminf_{n\to\infty} f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

LLMwiki: `Fatou's lemma; ID: durrett_ch1_fatou_integral`

Monotone Convergence Theorem

Theorem

If $0 \leq f_n \uparrow f$, then

$$\int f_n \, d\mu \uparrow \int f \, d\mu.$$

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Dominated Convergence Theorem

Theorem

If $f_n \to f$ a.e. and $|f_n| \leq g$ for one integrable g, then

$$\int f_n \, d\mu \to \int f \, d\mu.$$

LLMwiki: `Dominated convergence theorem; ID: durrett_ch1_dominated_convergence`

Exercise 1.5.1

Let

$$\|f\|_\infty = \inf\{M : \mu(\{x : |f(x)| > M\}) = 0\}.$$

Prove that

$$\int |fg| \, d\mu \leq \|f\|_1 \|g\|_\infty.$$

LLMwiki: `Essential supremum bounds integrals; ID: exercise_method_essential_supremum_bound`

Exercise 1.5.1: Proof Skeleton

LLMwiki: `Essential supremum bounds integrals; ID: exercise_method_essential_supremum_bound`

Exercise 1.5.2

Show that if μ is a probability measure, then

$$\|f\|_\infty = \lim_{p\to\infty} \|f\|_p.$$

LLMwiki: `Lp norms converge to Linfty on probability spaces; ID: exercise_method_lp_to_l_infty_limit`

Exercise 1.5.2: Proof Skeleton

LLMwiki: `Lp norms converge to Linfty on probability spaces; ID: exercise_method_lp_to_l_infty_limit`

Exercise 1.5.3

Prove Minkowski's inequality. For $p \in (1, \infty)$, use Holder's inequality to show

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Then show the same result for $p = 1$ and $p = \infty$.

LLMwiki: `Minkowski's inequality; ID: durrett_ch1_minkowski`

Exercise 1.5.3: Proof Skeleton

LLMwiki: `Minkowski's inequality; ID: durrett_ch1_minkowski`

Exercise 1.5.4

If f is integrable and E_m are disjoint sets with union E, show that

$$\sum_{m=0}^{\infty} \int_{E_m} f \, d\mu = \int_E f \, d\mu.$$

Conclude that if $f \geq 0$, then

$$\nu(E) = \int_E f \, d\mu$$

defines a measure.

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.4: Proof Skeleton

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.5

If $g_n \uparrow g$ and

$$\int g_1^- \, d\mu < \infty,$$

show that

$$\int g_n \, d\mu \uparrow \int g \, d\mu.$$

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.5: Proof Skeleton

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.6

If $g_m \geq 0$, show that

$$\int \sum_{m=0}^{\infty} g_m \, d\mu = \sum_{m=0}^{\infty} \int g_m \, d\mu.$$

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.6: Proof Skeleton

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.7

Let $f \geq 0$.

1. Show that
$$\int f \wedge n \, d\mu \uparrow \int f \, d\mu.$$
2. Use this to conclude that if g is integrable and $\varepsilon > 0$, then there is $\delta > 0$ such that $\mu(A) < \delta$ implies
$$\int_A |g| \, d\mu < \varepsilon.$$

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.7: Proof Skeleton

LLMwiki: `Monotone convergence theorem; ID: durrett_ch1_monotone_convergence`

Exercise 1.5.8

Show that if f is integrable on $[a, b]$ and

$$g(x) = \int_{[a,x]} f(y)\, dy,$$

then g is continuous on (a, b).

LLMwiki: `Absolute continuity of the integral; ID: durrett_ch1_dominated_convergence`

Exercise 1.5.8: Proof Skeleton

LLMwiki: `Absolute continuity of the integral; ID: durrett_ch1_dominated_convergence`

Exercise 1.5.9

Show that if

$$\|f\|_p = \left(\int |f|^p \, d\mu \right)^{1/p} < \infty,$$

then there are simple functions ϕ_n such that

$$\|\phi_n - f\|_p \to 0.$$

LLMwiki: `Simple functions are dense in Lp; ID: exercise_method_density_of_simple_functions_in_lp`

Exercise 1.5.9: Proof Skeleton

LLMwiki: `Simple functions are dense in Lp; ID: exercise_method_density_of_simple_functions_in_lp`

Exercise 1.5.10

Show that if

$$\sum_n \int |f_n| \, d\mu < \infty,$$

then

$$\sum_n \int f_n \, d\mu = \int \sum_n f_n \, d\mu.$$

LLMwiki: `Exchange integral and absolutely summable series; ID: exercise_method_absolute_summability_exchange`

Exercise 1.5.10: Proof Skeleton

LLMwiki: `Exchange integral and absolutely summable series; ID: exercise_method_absolute_summability_exchange`